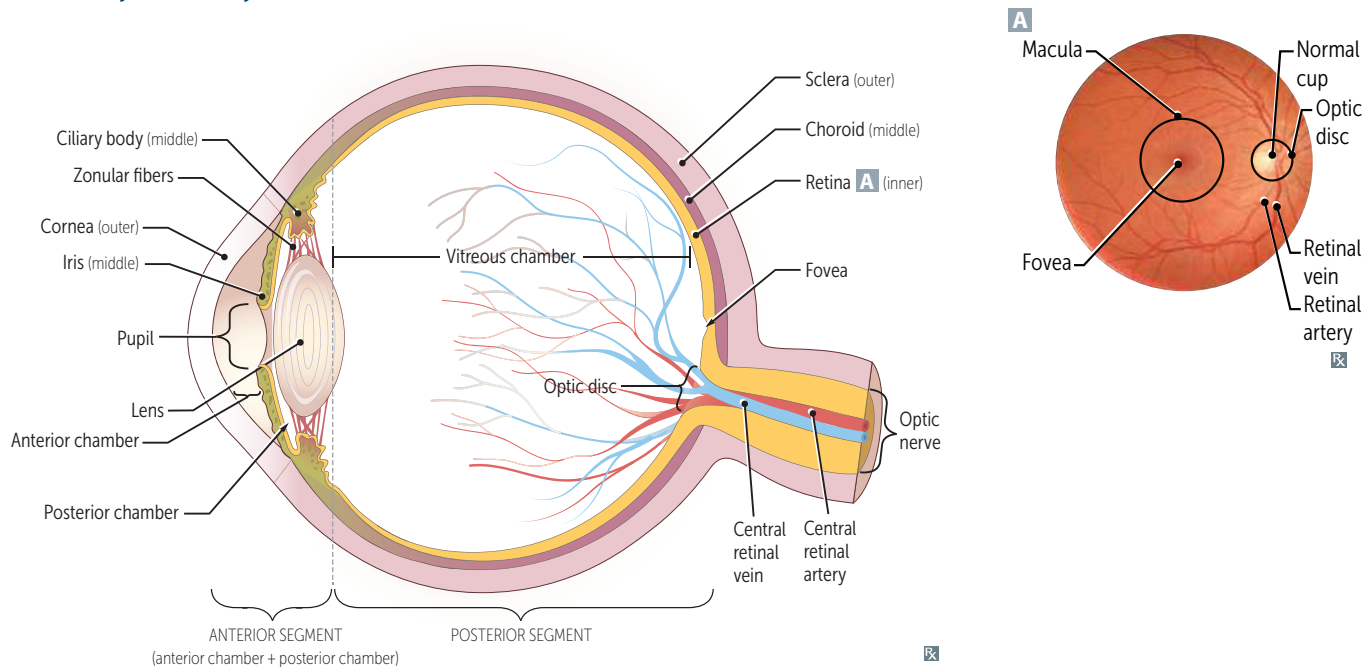


► NEUROLOGY—OPHTHALMOLOGY

Normal eye anatomy



Conjunctivitis



Inflammation of the conjunctiva → red eye **A**.

Allergic—itchy eyes, bilateral.

Bacterial—purulent discharge; treat with antibiotics.

Viral—most common, often adenovirus; sparse mucous discharge, swollen preauricular node, ↑ lacrimation; self-resolving.

Neonatal—eyelid swelling, exudative discharge after vaginal birth. Detect bacterial etiologies with NAAT. Maternal prenatal screening and treatment ↓ incidence.

- *C trachomatis* serotypes D-K (most common)—onset 5–14 days after birth; treat with erythromycin or azithromycin.
- *N gonorrhoeae*—onset 2–5 days after birth; prophylaxis: erythromycin eyedrops; treat with ceftriaxone to prevent blindness.
- Other pathogens—HSV, adenovirus.

Refractive errors

Common cause of impaired vision, correctable with glasses. **My cave** “hy”-des (hides) **vexed cylinders**.

Myopia

Most common. Also called “nearsightedness.” Eye too long for refractive power of cornea and lens → light focused in front of retina. Correct with **concave** (diverging) lens.

Hyperopia

Also called “farsightedness.” Eye too short for refractive power of cornea and lens → light focused behind retina. Correct with **convex** (converging) lens.

Astigmatism

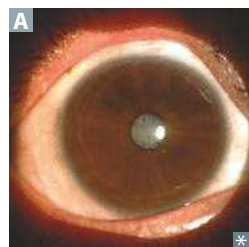
Irregular or asymmetric curvature of the cornea or lens → different refractive power at different axes. Correct with **cylindrical** lens.

Lens disorders

Presbyopia

Aging-related impairment in accommodation (focusing on near objects). Pathophysiology not fully understood but likely includes ↓ lens elasticity. Patients often need reading glasses or magnifiers.

Cataract



Painless, often bilateral, opacification of lens **A**. Can result in glare, loss of red reflex, and ↓ vision, especially at night.

Acquired risk factors: ↑ age, tobacco smoking, alcohol overuse, excessive sunlight, prolonged glucocorticoid use, diabetes mellitus, trauma, infection.

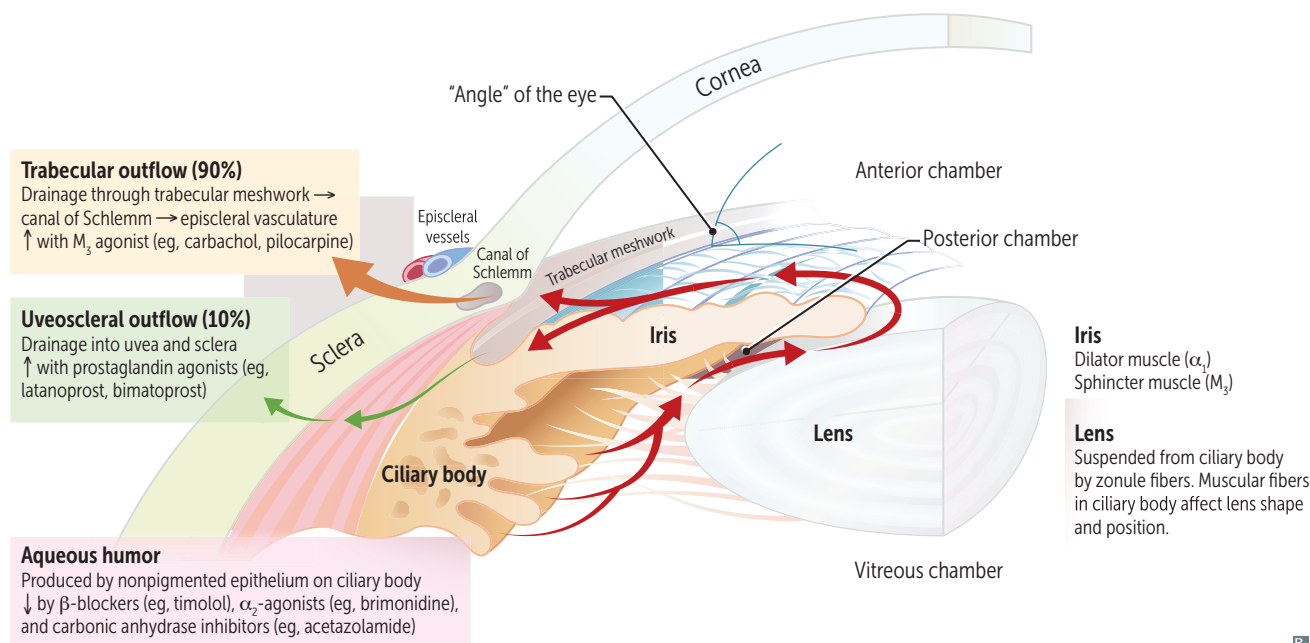
Congenital risk factors: classic galactosemia, galactokinase deficiency, trisomies (13, 18, 21), TORCH infections (eg, rubella), Marfan syndrome, Alport syndrome, myotonic dystrophy, NF-2.

Treatment: surgical replacement with an artificial lens.

Lens dislocation

Also called ectopia lentis. Displacement or malposition of lens. Usually due to trauma, but may occur in association with systemic diseases (eg, Marfan syndrome, homocystinuria).

Aqueous humor pathway



Glaucoma

Optic neuropathy causing progressive vision loss (peripheral → central). Usually, but not always, accompanied by ↑ intraocular pressure (IOP). Etiology is most often 1°, but can be 2° to an identifiable cause (eg, uveitis, glucocorticoids). Funduscopy: optic disc cupping (normal **A** vs thinning of outer rim of optic disc **B**). Treatment: pharmacologic or surgical lowering of IOP.

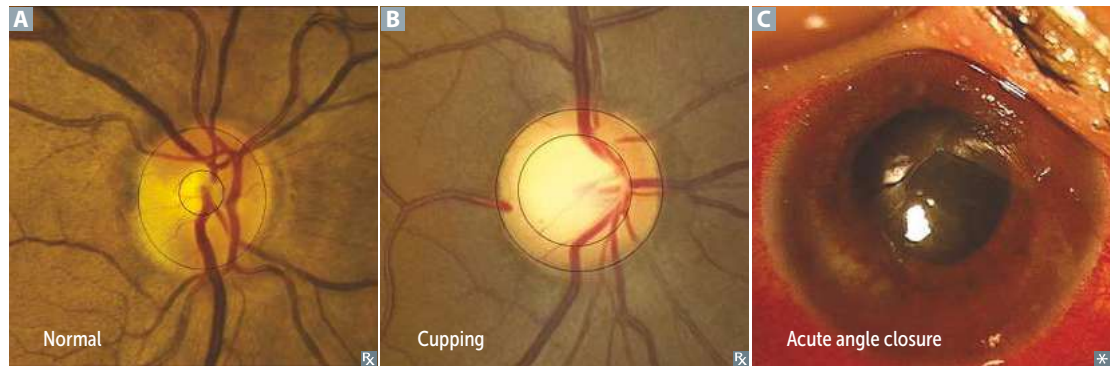
Open-angle glaucoma

Anterior chamber angle is open (normal). Most common type in US. Associated with ↑ resistance to aqueous humor drainage through trabecular meshwork. Risk factors: ↑ age, race (↑ incidence in Black population), family history, diabetes mellitus. Typically asymptomatic and discovered incidentally. Treat with prostaglandins or β-blockers.

Angle-closure glaucoma

Anterior chamber angle is narrowed or closed. Associated with anatomic abnormalities (eg, anteriorly displaced lens resting against central iris) → ↓ aqueous flow through pupil (pupillary block) → pressure buildup in posterior chamber → peripheral iris pushed against cornea → obstruction of drainage pathways by the iris. Usually chronic and asymptomatic, but may develop acutely.

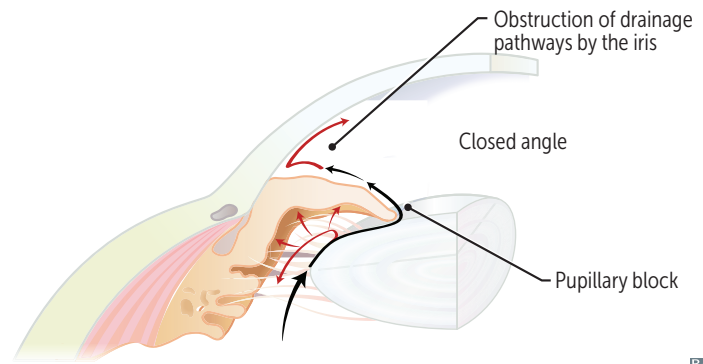
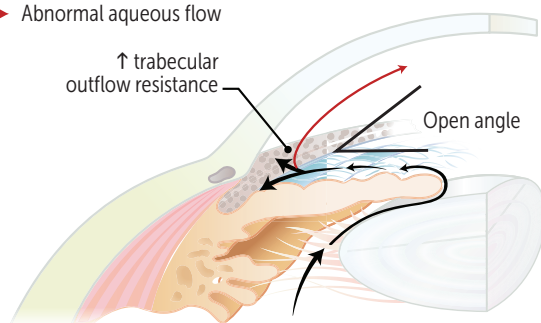
Acute angle-closure glaucoma—complete pupillary block causing abrupt angle closure and rapid ↑ IOP. Presents with severe eye pain, conjunctival erythema **C**, sudden vision loss, halos around lights, headache, fixed and mid-dilated pupil, nausea and vomiting. Hurts in a hurry with halos, a headache, and a “half-dilated” pupil. True ophthalmic emergency that requires immediate management to prevent blindness. Mydriatic agents are contraindicated. Treat with β-blocker, α₂-agonist, pilocarpine, acetazolamide, mannitol.



Open-angle glaucoma

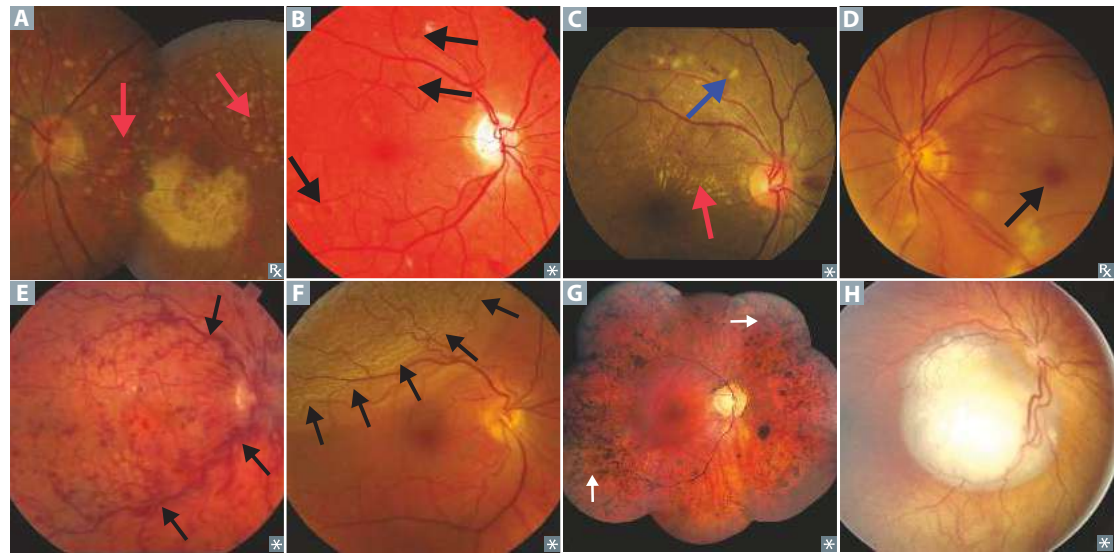
Angle-closure glaucoma

→ Normal aqueous flow
→ Abnormal aqueous flow

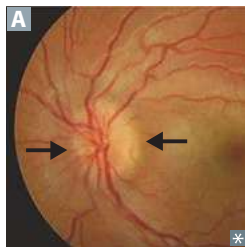


Retinal disorders

Age-related macular degeneration	Degeneration of macula (central area of retina) → loss of central vision (scotomas). Two types: ▪ Dry (most common)—gradual ↓ in vision with subretinal deposits (drusen, arrows in A). ▪ Wet—rapid ↓ in vision due to bleeding 2° to choroidal neovascularization. Distortion of straight lines (metamorphopsia) is an early symptom.
Diabetic retinopathy	Chronic hyperglycemia → ↑ permeability and occlusion of retinal vessels. Two types: ▪ Nonproliferative (most common)—microaneurysms, hemorrhages (arrows in B), cotton-wool spots, hard exudates. Vision loss mainly due to macular edema. ▪ Proliferative—retinal neovascularization due to chronic hypoxia. Abnormal new vessels may cause vitreous hemorrhage and tractional retinal detachment.
Hypertensive retinopathy	Chronic hypertension → spasm, sclerosis, and fibrinoid necrosis of retinal vessels. Funduscopy: arteriovenous nicking, microaneurysms, hemorrhages, cotton-wool spots (blue arrow in C), hard exudates (may form macular “star,” red arrow in C). Presence of papilledema is indicative of hypertensive emergency and warrants immediate lowering of blood pressure.
Retinal artery occlusion	Blockage of central or branch retinal artery usually due to embolism (carotid artery atherosclerosis > cardiogenic); less commonly due to giant cell arteritis. Presents with acute, painless monocular vision loss. Funduscopy: cloudy retina with “cherry-red” spot at fovea D, identifiable retinal emboli (eg, cholesterol crystals appear as small, yellow, refractile deposits in arterioles).
Retinal vein occlusion	Central retinal vein occlusion is due to 1° thrombosis; branch retinal vein occlusion is due to 2° thrombosis at arteriovenous crossings (sclerotic arteriole compresses adjacent venule causing turbulent blood flow). Funduscopy: retinal hemorrhage and venous engorgement (“blood and thunder” appearance; arrows in E), retinal edema in affected areas.
Retinal detachment	Separation of neurosensory retina from underlying retinal pigment epithelium → loss of choroidal blood supply → hypoxia and degeneration of photoreceptors. Two types: ▪ Rhegmatogenous (most common)—due to retinal tears; often associated with posterior vitreous detachment (↑ risk with advanced age, high myopia), less frequently traumatic. ▪ Nonrhegmatogenous—tractional or exudative (fluid accumulation). Commonly presents with symptoms of posterior vitreous detachment (eg, floaters, light flashes) followed by painless monocular vision loss (“dark curtain”). Funduscopy: opacification and wrinkling of detached retina F, change in vessel direction. Surgical emergency.
Retinitis pigmentosa	Group of inherited dystrophies causing progressive degeneration of photoreceptors and retinal pigment epithelium. May be associated with abetalipoproteinemia. Early symptoms: night blindness (nyctalopia) and peripheral vision loss. Funduscopy: triad of optic disc pallor, retinal vessel attenuation, and retinal pigmentation with bone spicule-shaped deposits G.
Retinopathy of prematurity	Preterm birth → loss of normal hypoxic environment in utero → relative hyperoxia (↑ with supplemental O₂ for NRDS) → ↓ VEGF → arrest of normal retinal vascularization. As the eyes grow → hypoxia of avascular retina → ↑ VEGF → retinal neovascularization (may cause tractional retinal detachment). Common cause of childhood blindness.
Retinoblastoma	Most common intraocular malignancy in children. Arises from immature retinal cells H. Caused by mutations to both <i>RBI</i> tumor suppressor genes on chromosome 13, which normally impede G₁ → S phase progression. Can be sporadic or familial (loss of heterozygosity). Presents with leukocoria, strabismus, nystagmus, eye redness.

Retinal disorders (continued)**Papilledema**

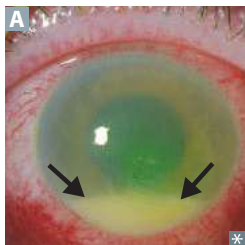
Optic disc swelling (usually bilateral) due to ↑ ICP (eg, 2° to mass effect). Results from impaired axoplasmic flow in optic nerve. Funduscopy: elevated optic disc with blurred margins **A**.

**Leukocoria**

Loss (whitening) of the red reflex. Important causes in children include retinoblastoma **A**, congenital cataract.

**Uveitis**

Inflammation of uvea; specific name based on location within affected eye. Anterior uveitis: iritis; posterior uveitis: choroiditis and/or retinitis. May have hypopyon (accumulation of pus in anterior chamber **A**) or conjunctival redness. Associated with systemic inflammatory disorders (eg, sarcoidosis, Behçet syndrome, juvenile idiopathic arthritis, HLA-B27-associated conditions).



Pupillary control**Miosis**

Constriction, parasympathetic:

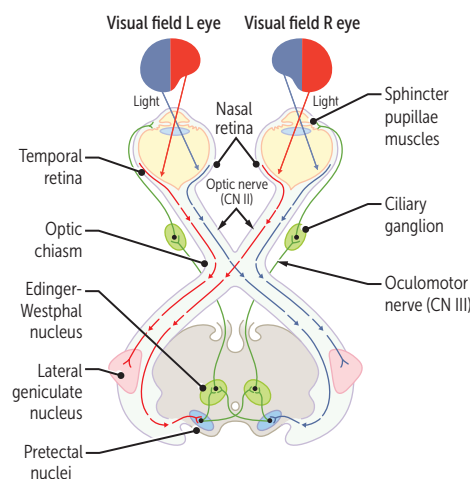
- 1st neuron: Edinger-Westphal nucleus to ciliary ganglion via CN III
- 2nd neuron: short ciliary nerves to sphincter pupillae muscles

Short ciliary nerves **shorten** the pupil diameter.

Pupillary light reflex

Light in either retina sends a signal via CN II to pretectal nuclei in midbrain that activates bilateral Edinger-Westphal nuclei; pupils constrict bilaterally (direct and consensual reflex).

Result: illumination of 1 eye results in bilateral pupillary constriction.

**Mydriasis**

Dilation, sympathetic:

- 1st neuron: hypothalamus to ciliospinal center of Budge (C8–T2)
- 2nd neuron: exit at T1 to superior cervical ganglion (travels along cervical sympathetic chain near lung apex, subclavian vessels)
- 3rd neuron: plexus along internal carotid, through cavernous sinus; enters orbit as long ciliary nerve to pupillary dilator muscles. Sympathetic fibers also innervate smooth muscle of eyelids (minor retractors) and sweat glands of forehead and face.

Long ciliary nerves make the pupil diameter **longer**.

Relative afferent pupillary defect

Also called Marcus Gunn pupil. Extent of pupillary constriction differs when light is shone in one eye at a time due to unilateral or asymmetric lesions of afferent limb of pupillary reflex (eg, retina, optic nerve). When light shines into a normal eye, constriction of the ipsilateral eye (direct reflex) and contralateral eye (consensual reflex) is observed. When light is swung from a normal eye to an affected eye, both pupils dilate instead of constricting.

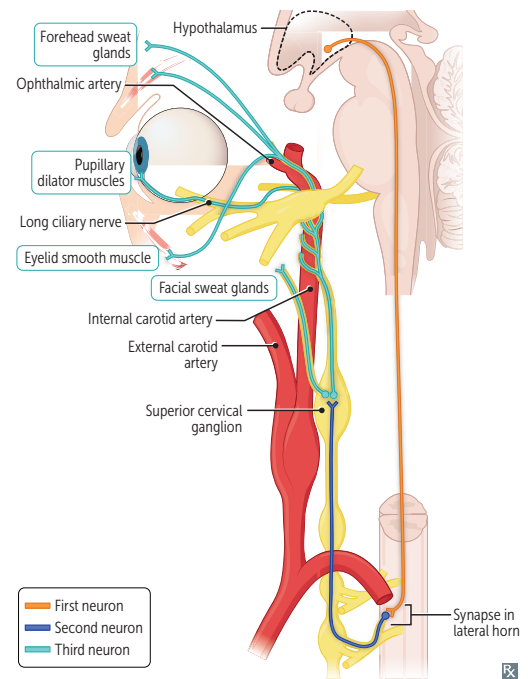
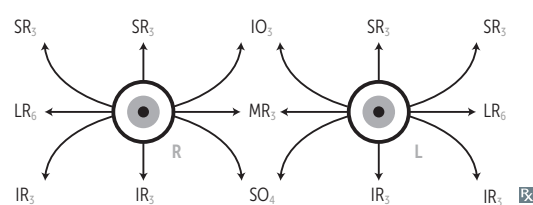
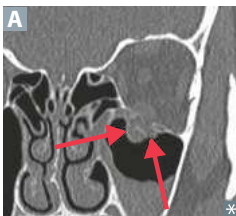
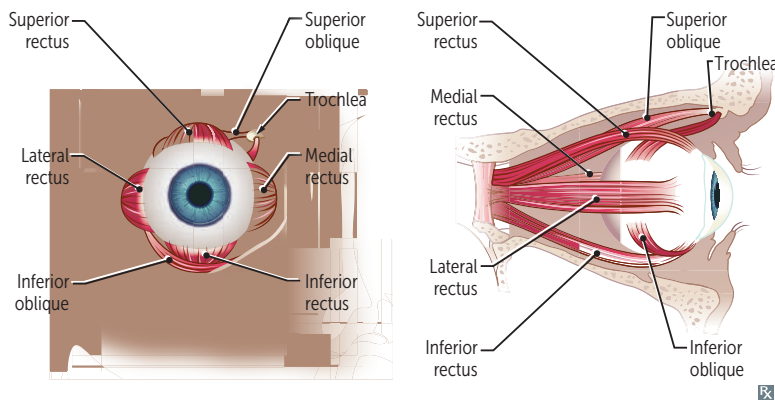
Horner syndrome

Sympathetic denervation of face:

- Ptosis (slight drooping of eyelid: superior tarsal muscle)
- Miosis (pupil constriction)
- Anhidrosis (absence of sweating) and absence of flushing of affected side of face

Associated with lesions along the sympathetic chain:

- 1st neuron: pontine hemorrhage, lateral medullary syndrome, spinal cord lesion above T1 (eg, Brown-Séquard syndrome, late-stage syringomyelia)
- 2nd neuron: stellate ganglion compression by Pancoast tumor
- 3rd neuron: carotid dissection (painful); anhidrosis is usually absent

**Ocular motility**

CN **VI** innervates the **L**ateral **R**ectus.
CN **IV** innervates the **S**uperior **O**blique.
CN **III** innervates the **R**est.
The “chemical formula” **LR₆SO₄R₅**.

Arrows in illustration depict direction of gaze with which to test each muscle.

Oblques go **O**pposite (left SO and IO tested with patient looking right)

IOU: **IO** tested looking **U**p

Blowout fracture—orbital floor fracture; usually due to trauma to eyeball or infraorbital rim. ↑ risk of IR muscle **A** and/or orbital fat entrapment. May lead to infraorbital nerve injury.

Strabismus

Eye misalignment (“crossed eyes”). Deviation of eye toward the nose (esotropia) is the most common type of strabismus in children. Complications include amblyopia, diplopia, adverse psychosocial impact.

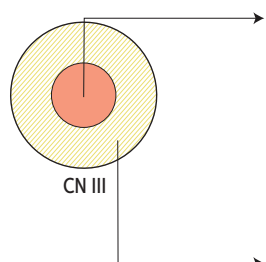
Amblyopia (“lazy eye”)—↓ visual acuity due to maldevelopment of visual cortex. Caused by abnormal visual experience early in life (eg, due to strabismus). Typically unilateral.

Cranial nerve III, IV, VI palsies

CN III damage

CN III has both motor (central) and parasympathetic (peripheral) components. Common causes include:

- Ischemia → pupil sparing (motor fibers affected more than parasympathetic fibers)
- Uncal herniation → coma
- PCom aneurysm → sudden-onset headache
- Cavernous sinus thrombosis → proptosis, involvement of CNs IV, V₁/V₂, VI
- Midbrain stroke → contralateral hemiplegia

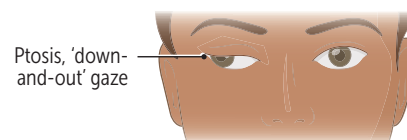


Motor output to extraocular muscles—affected primarily by vascular disease (eg, diabetes mellitus: glucose → sorbitol) due to ↓ diffusion of oxygen and nutrients to the interior (**m**iddle) fibers from compromised vasculature that resides on outside of nerve. Signs: ptosis, “down-and-out” gaze.

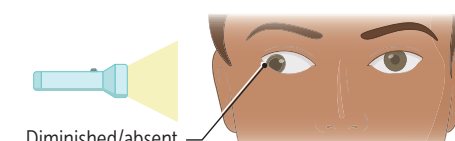
Parasympathetic output—fibers on the **p**eriphery are first affected by compression (eg, PCom aneurysm, uncal herniation). Signs: diminished or absent pupillary light reflex, “blown pupil” often with “down-and-out” gaze.

Motor = **m**iddle (central)

Parasympathetic = **p**eripheral



CN III (oculomotor) palsy, motor

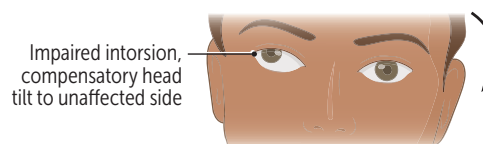


Diminished/absent pupillary light reflex, ‘blown pupil’, +/- ‘down-and-out’ gaze

CN III (oculomotor) palsy, parasympathetic

CN IV damage

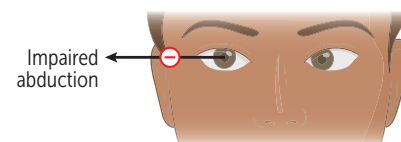
Pupil is higher in the affected eye. Characteristic head tilt to contralateral/unaffected side to compensate for lack of intorsion in affected eye. Can't see the **f**loor with CN **I**V damage (eg, difficulty going down stairs, reading).



CN IV (trochlear) palsy

CN VI damage

Affected eye unable to abduct and is displaced medially in primary position of gaze.



CN VI (abducens) palsy

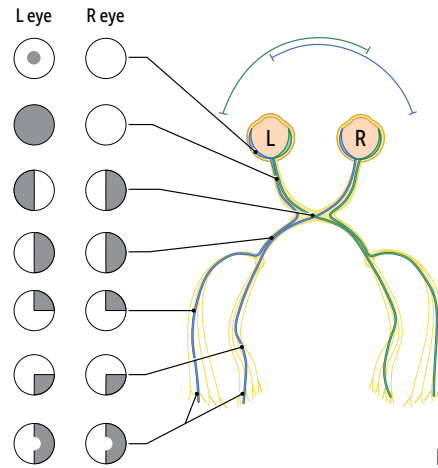
Visual field defects

Ventral optic radiation (Meyer **l**oop)—**l**ower retina; travels through temporal lobe; **l**oops around inferior horn of **l**ateral ventricle.

Dorsal optic radiation—superior retina; travels through parietal lobe.

Defect in visual field of:

- 1 **Macula**
Central scotoma
(*macular degeneration*)
- 2 **Optic nerve**
Left anopia
- 3 **Optic chiasm**
Bitemporal hemianopia
- 4 **Optic tract**
Right homonymous hemianopia
- 5 **Meyer loop**
Right upper quadrantanopia
(*left temporal lesion*)
- 6 **Dorsal optic radiation**
Right lower quadrantanopia
(*left parietal lesion*)
- 7 **Visual cortex**
Right homonymous hemianopia
with macular sparing
(*PCA infarct*)



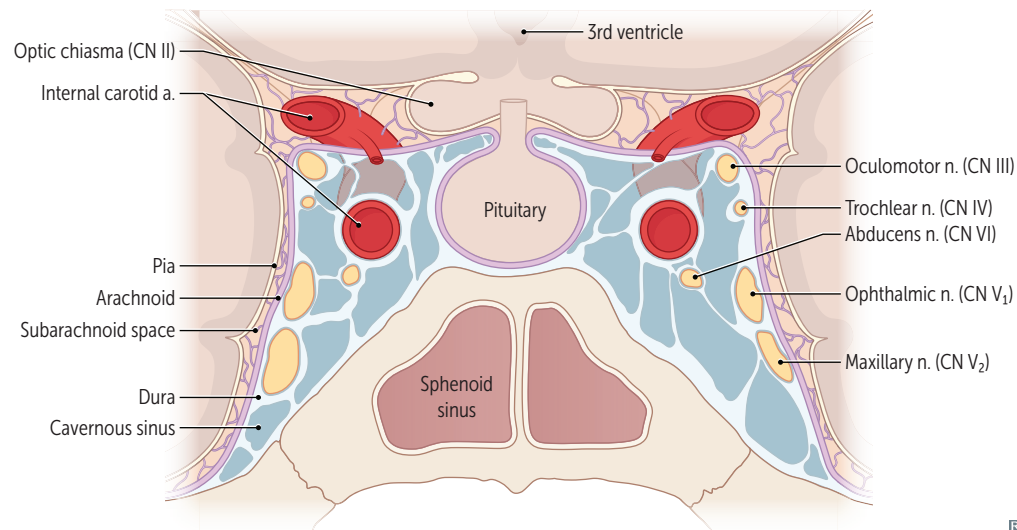
Note: When an image hits 1° visual cortex, it is upside down and left-right reversed.

Cavernous sinus

Collection of venous sinuses on either side of pituitary. Blood from eye and superficial cortex → cavernous sinus → internal jugular vein.

CNs III, IV, V₁, V₂, and VI plus postganglionic sympathetic pupillary fibers en route to orbit all pass through cavernous sinus. Cavernous portion of internal carotid artery is also here. **I**nternal carotid artery, **T**rigeminal nerve (**O**phthalmic and **M**axillary divisions), **A**bducens nerve, **T**rochlear nerve, **O**culomotor nerve (**I, TOMATO**).

Cavernous sinus syndrome—presents with variable ophthalmoplegia (eg, CN III and CN VI), ↓ corneal sensation, Horner syndrome and occasional ↓ maxillary sensation. 2° to pituitary tumor mass effect, carotid-cavernous fistula, or cavernous sinus thrombosis related to infection (spread due to lack of valves in dural venous sinuses).



Internuclear ophthalmoplegia

Medial longitudinal fasciculus (MLF): pair of tracts that interconnect CN VI and CN III nuclei. Coordinates both eyes to move in same horizontal direction. Highly myelinated (must communicate quickly so eyes move at same time). Lesions may be unilateral or bilateral (latter classically seen in multiple sclerosis, stroke).

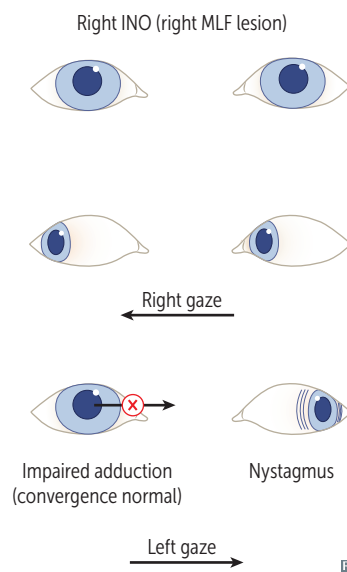
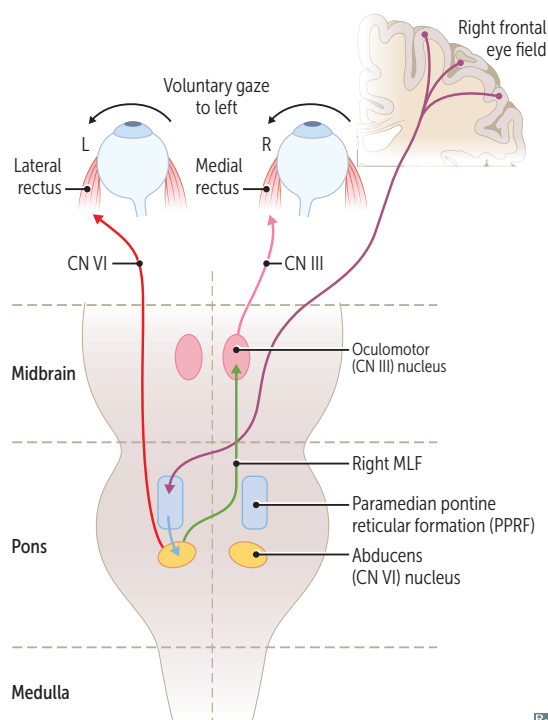
Lesion in MLF = internuclear ophthalmoplegia (INO), a conjugate horizontal gaze palsy. Lack of communication such that when CN VI nucleus activates ipsilateral lateral rectus, contralateral CN III nucleus does not stimulate medial rectus to contract. Abducting eye displays nystagmus (CN VI overfires to stimulate CN III). Convergence normal.

MLF in MS.

When looking left, the left nucleus of CN VI fires, which contracts the left lateral rectus and stimulates the contralateral (right) nucleus of CN III via the right MLF to contract the right medial rectus.

Directional term (eg, right INO, left INO) refers to the eye that is unable to adduct.

INO = **I**psilateral adduction failure, **N**ystagmus **O**pposite.



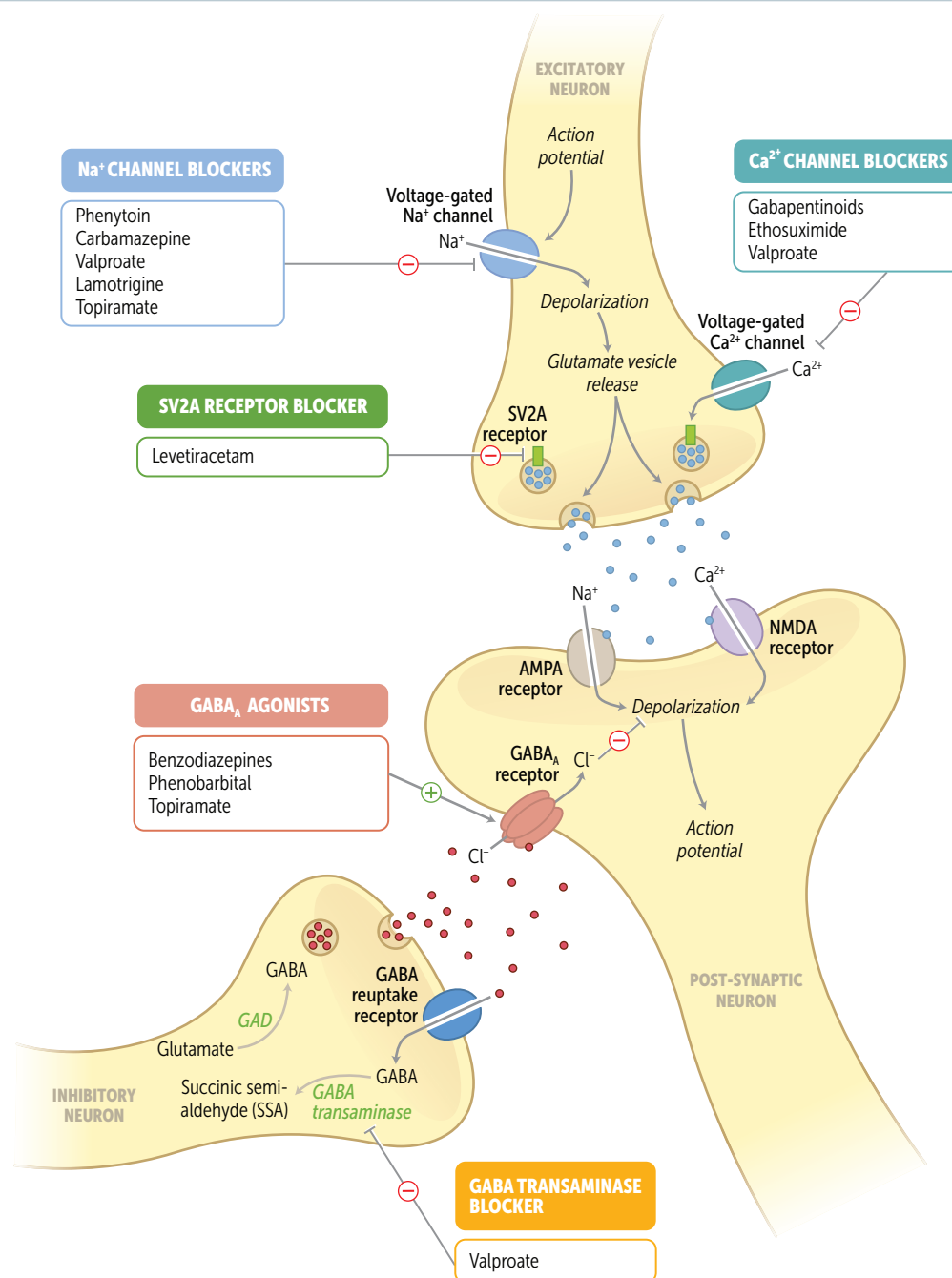
Eyelid disorders

DISORDER	PRESENTATION
Preseptal cellulitis	Anterior soft tissue eyelid infection. Mild presentation with unilateral ocular pain, swelling, and erythema present at rest.
Orbital cellulitis	Posterior eyelid infection affecting orbit contents (fat and muscles). Pain with ocular movement. Infection affecting the orbital contents (fat and muscles), usually secondary to bacterial sinusitis. Pain and double vision (diplopia) with ocular movement. Risk of vision loss, cavernous sinus thrombosis, and intracranial spread. Most commonly caused by <i>S aureus</i> and streptococci.
Blepharitis	Eyelid margin and lid inflammation, irritation, and crusting.
Hordeolum (stye)	Acute infection of the sebaceous or sweat glands of the eyelid. Tender, erythematous nodule.
Chalazion	Noninfectious granulomatous inflammation caused by obstruction of a meibomian (modified sebaceous) or Zeis (sebaceous) gland.
Xanthelasma	Yellowish patch on medial eyelid. May be associated with genetic and lifestyle factors, eg, high cholesterol.

► NEUROLOGY—PHARMACOLOGY

Anticonvulsants

	MECHANISM	COMMON ADVERSE EFFECTS	RARE BUT SERIOUS ADVERSE EFFECTS
Narrow spectrum (focal seizures)			
Phenytoin	Block Na ⁺ channel	Sedation, dizziness, diplopia, gingival hypertrophy (preventable with folate supplementation), rash, hirsutism, drug interactions (CYP450 induction)	SJS, DRESS, hepatotoxicity, neuropathy, osteoporosis, folate depletion, teratogenicity
Carbamazepine		Sedation, dizziness, diplopia, vomiting, diarrhea, SIADH, rash, drug interactions (CYP450 induction)	SJS, DRESS, hepatotoxicity, agranulocytosis, aplastic anemia, folate depletion, teratogenicity
Gabapentinoids Gabapentin, pregabalin	Block Ca ²⁺ channel	Sedation, dizziness, ataxia, weight gain	
Narrow spectrum (absence seizures only)			
Ethosuximide	Blocks Ca ²⁺ channel	Sedation, dizziness, vomiting	
Broad spectrum (focal and generalized seizures)			
Valproate	Blocks Na ⁺ channel Blocks Ca ²⁺ channel Blocks GABA transaminase	Sedation, dizziness, vomiting, weight gain, hair loss, easy bruising, drug interactions (CYP450 inhibition)	Hepatotoxicity, pancreatitis, teratogenicity (highest risk of all anticonvulsants)
Lamotrigine	Blocks Na ⁺ channel	Sedation, dizziness, rash	SJS, DRESS
Levetiracetam	Blocks Synaptic Vesicle protein 2A (SV2A)	Sedation, dizziness, fatigue	Neuropsychiatric (eg, psychosis)
Topiramate	Blocks Na ⁺ channel Potentiates GABA _A receptor	Sedation, dizziness, mood disturbance (eg, depression), weight loss, paresthesia	Kidney stones, angle-closure glaucoma

Anticonvulsants (*continued*)

F5

Barbiturates	Phenobarbital, pentobarbital.
MECHANISM	Facilitate GABA _A action by ↑ duration of Cl ⁻ channel opening, thus ↓ neuron firing (barbiturates ↑ duration).
CLINICAL USE	Sedative for anxiety, seizures, insomnia; alcohol or sedative withdrawal.
ADVERSE EFFECTS	Respiratory and cardiovascular depression (can be fatal); CNS depression (exacerbated by alcohol use); dependence; drug interactions (induces CYP-450). Contraindicated in porphyria. Overdose treatment is supportive (assist respiration and maintain BP).

Benzodiazepines

Diazepam, lorazepam, triazolam, temazepam, oxazepam, midazolam, chlordiazepoxide, alprazolam.

MECHANISM	Facilitate GABA _A action by ↑ frequency of Cl ⁻ channel opening (“frenzodiazepines” ↑ frequency). ↓ REM sleep. Most have long half-lives and active metabolites (exceptions [ATOM]: Alprazolam, Triazolam, Oxazepam, and Midazolam are short acting → higher addictive potential).
CLINICAL USE	Anxiety, panic disorder, spasticity, status epilepticus (lorazepam, diazepam, midazolam), eclampsia, medically supervised withdrawal (eg, alcohol/DTs; long-acting chlordiazepoxide and diazepam are preferred), night terrors/sleepwalking (↓ N3 and REM sleep), general anesthetic (amnesia, muscle relaxation), hypnotic (insomnia). Lorazepam, Oxazepam, and Temazepam can be used for those with liver disease who drink a LOT due to minimal first-pass metabolism.
ADVERSE EFFECTS	Dependence, additive CNS depression effects with alcohol and barbiturates (all bind the GABA _A receptor). Less risk of respiratory depression and coma than with barbiturates. Treat overdose with flumazenil (competitive antagonist at GABA benzodiazepine receptor). Can precipitate seizures by causing acute benzodiazepine withdrawal.

Insomnia therapy

AGENT	MECHANISM	ADVERSE EFFECTS	NOTES
Nonbenzodiazepine hypnotics	Examples: Zolpidem, Zaleplon, esZopiclone Act via the BZ ₁ subtype of GABA receptor	Ataxia, headaches, confusion Cause only modest day-after psychomotor depression and few amnestic effects (vs older sedative-hypnotics)	These ZZZs put you to sleep Short duration due to rapid metabolism by liver enzymes; effects reversed by flumazenil ↓ dependency risk and ↓ sleep cycle disturbance (vs benzodiazepine hypnotics)
Suvorexant	Orexin (hypocretin) receptor antagonist	CNS depression (somnia), headache, abnormal sleep-related activities	Contraindications: narcolepsy, combination with strong CYP3A4 inhibitors Not recommended in patients with liver disease Limited risk of dependency
Ramelteon	Melatonin receptor agonist: binds MT1 and MT2 in suprachiasmatic nucleus	Dizziness, nausea, fatigue, headache	No known risk of dependency

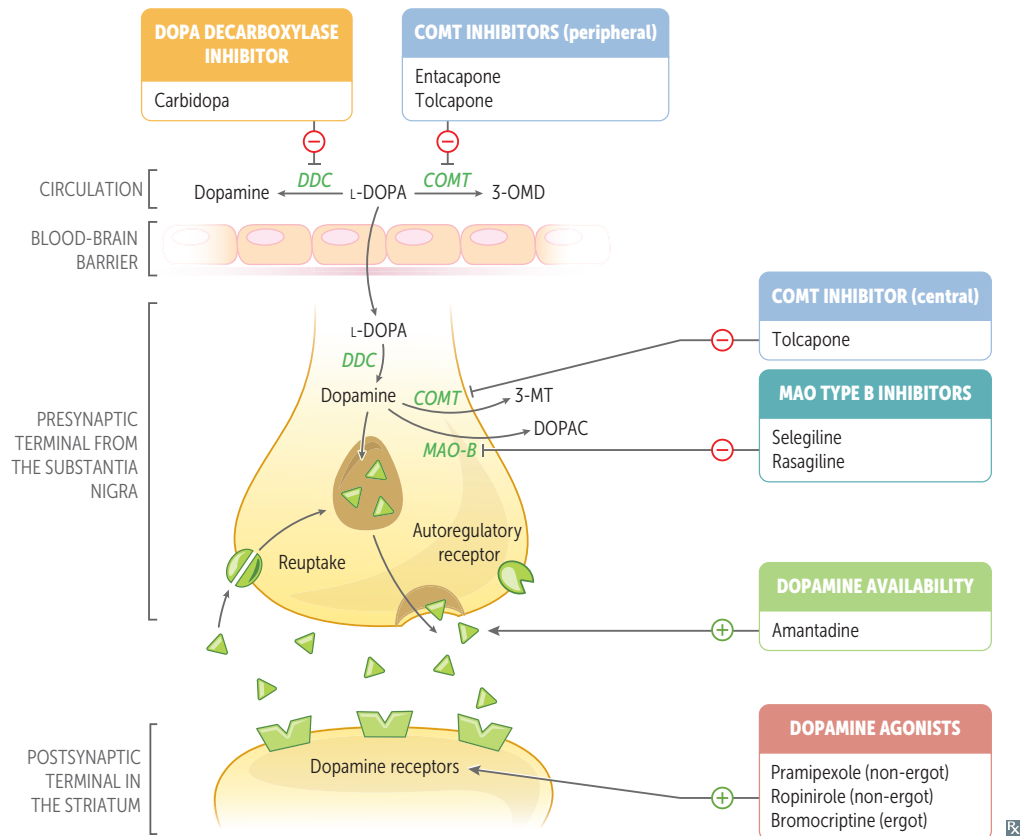
Triptans**Sumatriptan**

MECHANISM	5-HT _{1B/1D} agonists. Inhibit trigeminal nerve activation, prevent vasoactive peptide release, induce vasoconstriction.
CLINICAL USE	Acute migraine, cluster headache attacks. A sumo wrestler trips and falls on their head.
ADVERSE EFFECTS	Coronary vasospasm (contraindicated in patients with CAD or vasospastic angina), mild paresthesia, serotonin syndrome (in combination with other 5-HT agonists).

Parkinson disease therapy

Most effective treatments are non-ergot dopamine agonists (younger patients) and levodopa/carbidopa (older patients). Deep brain stimulation of the STN or GPi aids in advanced disease.

STRATEGY	AGENTS
Dopamine agonists	Non-ergot (preferred)—pramipexole, ropinirole; toxicity includes nausea, impulse control disorder (eg, gambling), postural hypotension, hallucinations, confusion, sleepiness, edema. Ergot—bromocriptine; rarely used due to toxicity.
↑ dopamine availability	Amantadine (↑ dopamine release and ↓ dopamine reuptake); mainly used to reduce levodopa-induced dyskinesias; toxicity = peripheral edema, livedo reticularis, ataxia.
↑ L-DOPA availability	Agents prevent peripheral (pre-BBB) L-DOPA degradation → ↑ L-DOPA entering CNS → ↑ central L-DOPA available for conversion to dopamine. - Levodopa (L-DOPA)/carbidopa—carbidopa blocks peripheral conversion of L-DOPA to dopamine by inhibiting DOPA decarboxylase. Also reduces adverse effects of peripheral L-DOPA conversion into dopamine (eg, nausea, vomiting). - Entacapone and tolcapone prevent peripheral L-DOPA degradation to 3-O-methyldopa (3-OMD) by inhibiting COMT. Used in conjunction with levodopa.
Prevent dopamine breakdown	Agents act centrally (post-BBB) to inhibit breakdown of dopamine. Selegiline, rasagiline—block dopamine → 3,4-dihydroxyphenylacetic acid (DOPAC) conversion via selective inhibition of MAO-B, which is more commonly found in the Brain than periphery. Improve symptoms for patients with cyclic fluctuations in levodopa efficacy (on/off phenomenon). - Tolcapone—crosses BBB and blocks conversion of dopamine to 3-methoxytyramine (3-MT) in the brain by inhibiting central COMT. Tolcapone pays the toll and can cross the BBB.
Curb excess cholinergic activity	Benz tropine, tri hexyphenidyl (Antimuscarinic; improves tremor and rigidity but has little effect on bradykinesia in Parkinson disease). Tri Parking my Mercedes- Benz .



Carbidopa/levodopa

MECHANISM	↑ dopamine in brain. Unlike dopamine, L-DOPA can cross BBB and is converted by DOPA decarboxylase in the CNS to dopamine. Carbidopa, a peripheral DOPA decarboxylase inhibitor that cannot cross BBB, is given with L-DOPA to ↑ bioavailability of L-DOPA in the brain and to limit peripheral adverse effects.
CLINICAL USE	Parkinson disease.
ADVERSE EFFECTS	Nausea, hallucinations, postural hypotension. With progressive disease, L-DOPA can lead to “on-off” phenomenon with improved mobility during “on” periods, then impaired motor function during “off” periods when patient responds poorly to L-DOPA or medication wears off.

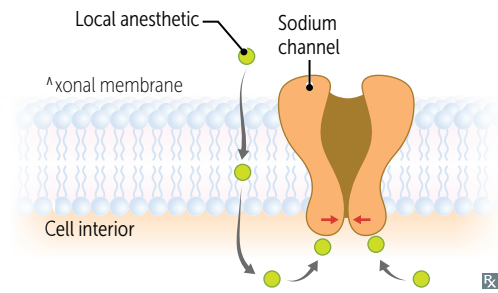
Neurodegenerative disease therapy

DISEASE	AGENT	MECHANISM	NOTES
Alzheimer disease	Donepezil, rivastigmine, galantamine	AChE inhibitor	1st-line treatment Adverse effects: nausea, dizziness, insomnia; contraindicated in patients with cardiac conduction abnormalities Dona Riva forgot the gala
	Memantine	NMDA receptor antagonist; helps prevent excitotoxicity (mediated by Ca^{2+})	Used for moderate to advanced dementia Adverse effects: dizziness, confusion, hallucinations
Amyotrophic lateral sclerosis	Riluzole	↓ neuron glutamate excitotoxicity	↑ survival Treat Lou Gehrig disease with riLou zole
Huntington disease	Deutetrabenazine, tetrabenazine	Inhibit vesicular monoamine transporter (VMAT) → ↓ dopamine vesicle packaging and release	May be used for Huntington chorea and tardive dyskinesia

Local anesthetics

Esters—benzocaine, chlorprocaine, cocaine, tetracaine.

Amides—bupivacaine, lidocaine, mepivacaine, priloca~~i~~ne, ropivacaine (amides have 2 **i**'s in name).



MECHANISM	Block neurotransmission via binding to voltage-gated Na⁺ channels on inner portion of the channel along nerve fibers. Most effective in rapidly firing neurons. 3° amine local anesthetics penetrate membrane in uncharged form, then bind to ion channels as charged form. Can be given with vasoconstrictors (usually epinephrine) to enhance block duration of action by ↓ systemic absorption. In infected (acidic) tissue, alkaline anesthetics are charged and cannot penetrate membrane effectively → need more anesthetic. Order of loss: (1) pain, (2) temperature, (3) touch, (4) pressure.
CLINICAL USE	Minor surgical procedures, spinal anesthesia. If allergic to esters, give amides.
ADVERSE EFFECTS	CNS excitation, severe cardiovascular toxicity (bupivacaine), hypertension, hypotension, arrhythmias (cocaine), methemoglobinemia (benzocaine, priloca i ne).

General anesthetics

CNS drugs must be lipid soluble (cross the BBB) or be actively transported.

Drugs with ↓ solubility in blood (eg, nitrous oxide [N₂O]) = rapid induction and recovery times.

Drugs with ↑ solubility in lipids (eg, isoflurane) = ↑ potency.

MAC = **M**inimum **A**lveolar **C**oncentration (of inhaled anesthetic) required to prevent 50% of subjects from moving in response to noxious stimulus (eg, skin incision). Potency = 1/MAC.

MECHANISM		ADVERSE EFFECTS/NOTES
Inhaled anesthetics		
Sevoflurane	Mechanism unknown	Respiratory depression, ↓ cough reflex
Desflurane		Myocardial depression, ↓ BP
Isoflurane		↑ cerebral blood flow (↑ ICP), ↓ metabolic rate
N₂O		↓ skeletal and smooth muscle tone
		Postoperative nausea and vomiting
		Malignant hyperthermia
		Diffusion into and expansion (N ₂ O) of gas-filled cavities (eg, pneumothorax); very low potency
Intravenous anesthetics		
Propofol	Potentiates GABA _A receptor Inhibits NMDA receptor	Respiratory depression, ↓ BP; most commonly used IV agent for induction of anesthesia
Etomidate	Potentiates GABA _A receptor	Acute adrenal insufficiency, postoperative nausea and vomiting; hemodynamically neutral
Ketamine	Inhibits NMDA receptor	Sympathomimetic: ↑ BP, ↑ HR, ↑ cerebral blood flow (↑ ICP), bronchodilation Psychotomimetic: hallucinations, vivid dreams

Neuromuscular blocking drugs	Muscle paralysis in surgery or mechanical ventilation. Selective for N_m nicotinic receptors at neuromuscular junction but not autonomic N_n receptors.
Depolarizing neuromuscular blocking drugs	Succinylcholine—strong N_m nicotinic receptor agonist; produces sustained depolarization and prevents muscle contraction. Reversal of blockade: - Phase I (prolonged depolarization)—no antidote. Block potentiated by cholinesterase inhibitors. - Phase II (repolarized but blocked; N_m nicotinic receptors are available, but desensitized)—may be reversed with cholinesterase inhibitors. Complications include hypercalcemia, hyperkalemia, malignant hyperthermia. ↑ risk of prolonged muscle paralysis in patients with pseudocholinesterase deficiency.
Nondepolarizing neuromuscular blocking drugs	Atracurium, cisatracurium, pancuronium, rocuronium, vecuronium—competitive N_m nicotinic receptor antagonist. Reversal of blockade—sugammadex or cholinesterase inhibitors (eg, neostigmine). Anticholinergics (eg, atropine, glycopyrrolate) are given with cholinesterase inhibitors to prevent muscarinic effects (eg, bradycardia).
Malignant hyperthermia	Rare, life-threatening, hypermetabolic condition caused by the administration of potent inhaled anesthetics (sevoflurane, desflurane, isoflurane) or succinylcholine in susceptible individuals. Susceptibility to malignant hyperthermia is caused by de novo or inherited (autosomal dominant) mutations to ryanodine (<i>RYR1</i>) or dihydropyridine receptors (<i>DHPR</i>). ↑ ↑ Ca^{2+} release from sarcoplasmic reticulum → sustained muscle contraction → hypercapnia, tachycardia, masseter/generalized muscle rigidity, rhabdomyolysis, hyperthermia. Treatment: dantrolene (ryanodine receptor antagonist).

Skeletal muscle relaxants

DRUG	MECHANISM	CLINICAL USE	NOTES
Baclofen	GABA _B receptor agonist in spinal cord	Muscle spasticity, dystonia, multiple sclerosis	Acts on the back (spinal cord) May cause sedation
Cyclobenzaprine	Acts within CNS, mainly at the brainstem	Muscle spasms	C entrally acting Structurally related to TCAs May cause anticholinergic adverse effects, sedation
Dantrolene	Prevents release of Ca ²⁺ from sarcoplasmic reticulum of skeletal muscle by inhibiting the ryanodine receptor	Malignant hyperthermia (toxicity of inhaled anesthetics and succinylcholine) and neuroleptic malignant syndrome (toxicity of antipsychotics)	Acts d irectly on muscle
Tizanidine	α ₂ agonist, acts centrally	Muscle spasticity, multiple sclerosis, ALS, cerebral palsy	

Opioid analgesics

MECHANISM	Act as agonists at opioid receptors (μ = β-endorphin, δ = enkephalin, κ = dynorphin) to modulate synaptic transmission—close presynaptic Ca ²⁺ channels, open postsynaptic K ⁺ channels → ↓ synaptic transmission. Inhibit release of ACh, norepinephrine, 5-HT, glutamate, substance P.
EFFICACY	Full agonist: morphine, meperidine (long acting), methadone, codeine (prodrug; activated by CYP2D6), fentanyl. Partial agonist: buprenorphine. Mixed agonist/antagonist: butorphanol, nalbuphine. Antagonist: naloxone, naltrexone, methylnaltrexone.
CLINICAL USE	Moderate to severe or refractory pain, diarrhea (loperamide, diphenoxylate), acute pulmonary edema, maintenance programs for opiate use disorder (methadone, buprenorphine + naloxone), neonatal abstinence syndrome (methadone, morphine).
ADVERSE EFFECTS	Nausea, vomiting, pruritus (histamine release), opiate use disorder, respiratory depression, constipation, sphincter of Oddi spasm, miosis (except meperidine → mydriasis), additive CNS depression with other drugs. Tolerance does not develop to miosis and constipation. Treat toxicity with naloxone and prevent relapse with naltrexone once detoxified.

Tramadol

MECHANISM	Very weak opioid agonist; also inhibits the reuptake of norepinephrine and serotonin.
CLINICAL USE	Chronic pain.
ADVERSE EFFECTS	Similar to opioids; decreases seizure threshold; serotonin syndrome.

Butorphanol, nalbuphine

MECHANISM	μ -opioid receptor partial agonists and κ -opioid receptor full agonists.
CLINICAL USE	Analgesia for severe pain (eg, labor).
NOTES	Mixed opioid agonists/antagonists cause less respiratory depression than full opioid agonists. Can cause opioid withdrawal symptoms if patient is also taking full opioid agonist (due to competition for opioid receptors). Not easily reversed with naloxone.

Capsaicin

Naturally found in hot peppers.

MECHANISM	Excessive stimulation and desensitization of nociceptive fibers $\rightarrow$ $\downarrow$ substance P release $\rightarrow$ $\downarrow$ pain.
CLINICAL USE	Musculoskeletal and neuropathic pain.

Glaucoma therapy

$\downarrow$ IOP via $\downarrow$ amount of aqueous humor (inhibit synthesis/secretion or $\uparrow$ drainage).
 “ $\beta\alpha$ D humor may not be politically correct.”

DRUG CLASS	EXAMPLES	MECHANISM	ADVERSE EFFECTS
β-blockers	Timolol, betaxolol, carteolol	$\downarrow$ aqueous humor synthesis	No pupillary or vision changes
α-agonists	Epinephrine (α_1), apraclonidine, brimonidine (α_2)	$\downarrow$ aqueous humor synthesis via vasoconstriction (epinephrine) $\downarrow$ aqueous humor synthesis (apraclonidine, brimonidine) $\uparrow$ outflow of aqueous humor via uveoscleral pathway	Mydriasis (α_1); do not use in closed-angle glaucoma Blurry vision, ocular hyperemia, foreign body sensation, ocular allergic reactions, ocular pruritus
Diuretics	Acetazolamide	$\downarrow$ aqueous humor synthesis via inhibition of carbonic anhydrase	No pupillary or vision changes
Prostaglandins	Bimatoprost, latanoprost (PGF _{2α})	$\uparrow$ outflow of aqueous humor via $\downarrow$ resistance of flow through uveoscleral pathway	Darkens color of iris (browning), eyelash growth
Cholinomimetics (M₃)	Direct: pilocarpine, carbachol Indirect: physostigmine, echothiophate	$\uparrow$ outflow of aqueous humor via contraction of ciliary muscle and opening of trabecular meshwork Use pilocarpine in acute angle closure glaucoma—very effective at opening meshwork into canal of Schlemm	Miosis (contraction of pupillary sphincter muscles) and cyclospasm (contraction of ciliary muscle)